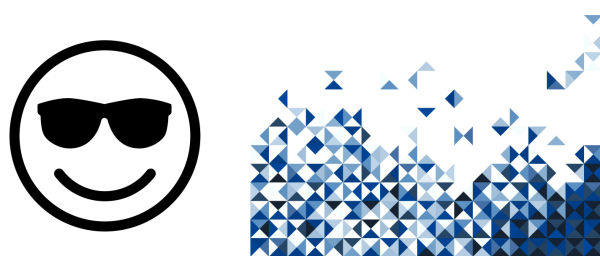


Graphical Abstract

Informational Content Analysis of Lost Languages

Richard Witt, Anamaria Berea



Template Figure Placeholder

Highlights

Informational Content Analysis of Lost Languages

Richard Witt, Anamaria Berea

- Bullet 1
- Bullet 2
- Bullet 3

Informational Content Analysis of Lost Languages

Richard Witt^{a,*}, Anamaria Berea^b

^a*Computational and Data Sciences, PhD Student, George Mason University, 4400
University Drive, Fairfax, 22030, Virginia, USA*

^b*Computational Social Sciences, Associate Professor, George Mason University, 4400
University Drive, Fairfax, 22030, Virginia, USA*

Abstract

This study investigates what information can be recovered from symbolic data such as lost languages. It compares two exploratory approaches that require no prior knowledge of the data. Machine learning clustering and information theoretic measurements. HDBSCAN is used to uncover recurring groups without predefined categories. Information theoretic measures quantify redundancy, uncertainty, and unicity distance.

Keywords: Keyword1, keyword2, keyword3

TODO: Update for new article

PACS: 01.78.+p, 89.20.Dd, 01.75.+m, 89.70.Cf

TODO: Update for new article

2000 MSC: 68T50, 91B82, 62P12, 91F99

1. Introduction

Deciphering Rongorongo, Linear A, and the Indus Valley Script would increase knowledge and understanding in archeology, anthropology, history, linguistics, astronomy and more]]. Still, research has been unable to yield satisfactory decipherment of these lost scripts and basic questions such as Rongorongo's status as a writing system is still an open question]]. Barthel created the now canonical transliterations for Rongorongo in 1958]]. Transliterations for Linear A and Indus Valley were provided by TODO in TODO]] and TODO in TODO]] respectively.

*Corresponding author

Email address: `rwitt4@gmu.edu` (Richard Witt)

Current research has yielded ... TODO 1-2 sentence summary.

This paper uses these transliterations to conduct an analysis using Natural Language Processing to provide insight on two different open questions. We first determine if there is enough informational content in the corpora for theoretical decipherment to be possible. Next we provide insight on the type of writing system for each lost language.

The remainder of this paper is structured as follows: Section 2 describes initial and current efforts to decipher the lost scripts. Section 3 details the source transliterations used in the analysis. Section 4 describes the methodology and rationale used to determine decipherability and writing system. Section 5 provides the analysis and results. Finally Section 6 discusses the results and describes potential paths for future work.

2. Literature Review

2.1. Easter Island and Rongo Rongo

Efforts to translate and decipher have spanned centuries.

Rapa Nui is still the spoken language of the indigenous people of Easter Island. []

Father Sebastian Englebert creates Rapa Nui to Spanish Dictionary in 1938, translated to spanish in and updated in 1978.[1]

Rapa Nui to Spanish []

In his 1958 work *Grundlagen zur Entzifferung der Osterinselschrift* Barthel created the alphanumeric cataloging system and transliterations from glyph to alpha numeric characters [2].

Horley provides corrected glyphs and

Jacques Guy provided the tablet detail [3] and original web resources Philip Spaelti used to create Kohaumotu.org which provides the glyph library this paper uses as input data. [4]

2.2.

3. Data

Two distinct data sets were used for this paper. One for Rongo Rongo and one for Linear A. Rongo Rongo is the extinct script of Easter Island. Linear A is also lost script that was used on Crete during the second millennium BCE.

3.1. *Rongo Rongo*

Data available at the [Rongorongo corpus \(codes only\)](#) was used for all analysis of Rongo Rongo. This site relies on the work initially recorded by Barthel in 1958 [1] which has been assembled into an impressive interactive website greatly aiding in the understanding of Rongo Rongo glyphs. This paper relies on the accuracy of transliterations from this prior work.

The Rongo Rongo corpus consists of 26 Tablets Coded A through Z and retrieved from [the Kohaumotu Rongorongo codes page](#). The updated 3 character glyph codes were used to create an encoding library of all encodings. Instead of using the glyphs themselves we rely directly on the index created by Barthel to represent a manual encoding of glyphs.

The glyphs are numbered in a systematic manner. Each row is intended to categorize the glyphs with Row 0 containing the simplest forms while higher rows increase in complexity. While only 90 rows are shown 799 total rows were categorized by Barthel and available in [1] and in the Kohaumotu.org website. [2].

To create the index used in this analysis the row and column are concatenated to result in the three digit index used. In this manner 000 represents the blank space in the upper left, and 001 represents both variants



Figure 1: **Rongo Rongo Great Washington Tablet.** Images created by the Smithsonian and compiled by .

	0	1	2	3	4	5	6	7	8	9
0										
10										
20										
30										
40										
50										
60										
70										
80										
90										

Figure 2: **Rongo Rongo Glyph Examples.** TODO.

3.2. Linear A

4. Methodology and rationale

We apply a machine learning clustering pipeline centered around an HDBSCAN model and information-theoretic techniques as complementary methods. HDBSCAN attempts to group data points into separable clusters by evaluating multivariate similarity among informational features. HDBSCAN is especially useful in unknown languages since it creates separation without requiring labels or a predetermined number of clusters. Information-theoretic measures summarize corpus properties such as uncertainty, redundancy, and properties relevant to decipherability. Shannon entropy measure the uncertainty in the distribution while redundancy reflects predictable or repeatable structure. Both approaches distinguish patterns that can be inferred from the data without linguistic or archeological interpretations.

frame the paper as a comparative analysis between ML specific methods (HDBSCAN) and information theoretic (and statistical) methods and compare.

It doesn't matter if these languages are logographic or syllabic, or alien. What I want you to focus on is on what kind of information we can extract

from data that we don't know too much about (is agnostic). Assume these are dolphin languages or alien languages.

4.1. Machine Learning with HDBSCAN

HDBSCAN pipeline consists of seven separate stages. Corpus Ingestion, TF-IDF Vectorization and UMAP Dimensionality Reduction prepare the data for the HDBSCAN algorithm. The HDBSCAN algorithm analyzes the data to form groups by evaluating the relative density and proximity of the data points. Topic extraction supports interpretation of resulting clusters and 3D Projection prepares the HDBSCAN output for 3 dimensional visualization. Each step of the process is described in the following paragraphs.

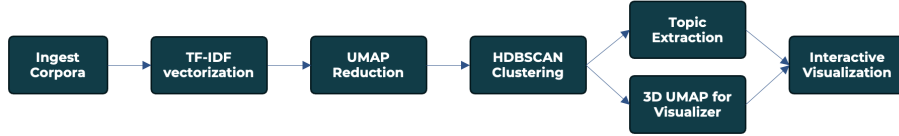


Figure 3: **HDBSCAN Pipeline.** HDBSCAN pipeline showing Corpus Ingestion, TF-IDF Vectorization, UMAP Dimensionality Reduction, HDBSCAN Clustering, Topic Extraction, Three Dimensional Projection, and Interactive Visualization used in the analysis.

4.2. Information Theoretical

Entropy Unicity Distance

4.3. Statistical

TODO

5. Analysis and Results

TODO

6. Discussion and Future Work

The methods in this paper cannot decipher the meaning or linguistic composition of an unknown symbolic system. Instead they determine if the available data contain non-random regularities and what scale those regularities happen. It can also be used to determine if the corpus is of sufficient size and non-random enough to support more specific analysis.

Appendix A. TODO

Year	Title	TODO
2001	Astronomy and Astrophysics for the New Millennium	Astrophysics
2003	The Sun to the Earth—and Beyond: A Decadal Research Strategy in Solar and Space Physics	Heliophysics
2003	New Frontiers in the Solar System: An Integrated Exploration Strategy	Planetary Science
2007	Earth Science and Applications from Space: National Imperatives for the Next Decade and Beyond	Earth Science

Table A.1: Decadal Surveys Used in Analysis

Appendix B. TODO2

This section details TODO

TODO1: Follow a `yyyy-DS-Category-####.pdf` format where `####` is the National Academies document number. `DS` is used to categorize all Decadal Surveys. `MA` replaces `DS` for Midterm Assessments, and `SP` is used for the Special Publication in 2015 that identified lessons learned.

TODO2: Follow a `yyyy_SSP_Nasa_File_Name.pdf` format. `SSP` represents Science mission directorate Science Plans. This is replaced with `AD` to indicate Additional Documentation, `NSP` to indicate NASA Science Plan, or `AD` to indicate additional documentation present on the SMD website.

References

- [1] S. Englert, Diccionario Rapanui-Español, 1948, translated from Spanish to English by Jacques Guy.
- [2] T. S. Barthel, Grundlagen zur Entzifferung der Osterinselschrift, Cram, de Gruyter, Hamburg, 1958.

- [3] J. Guy, [Rongorongo or the hieroglyphs of the easter island tablets](http://kohaumotu.org/rongorongo_org/), archived and mirrored by Philip Spaelti (2005).
URL http://kohaumotu.org/rongorongo_org/
- [4] P. Spaelti, [Glyph library](http://kohaumotu.org/Rongorongo/Glyph_Library/index.html).
URL http://kohaumotu.org/Rongorongo/Glyph_Library/index.html